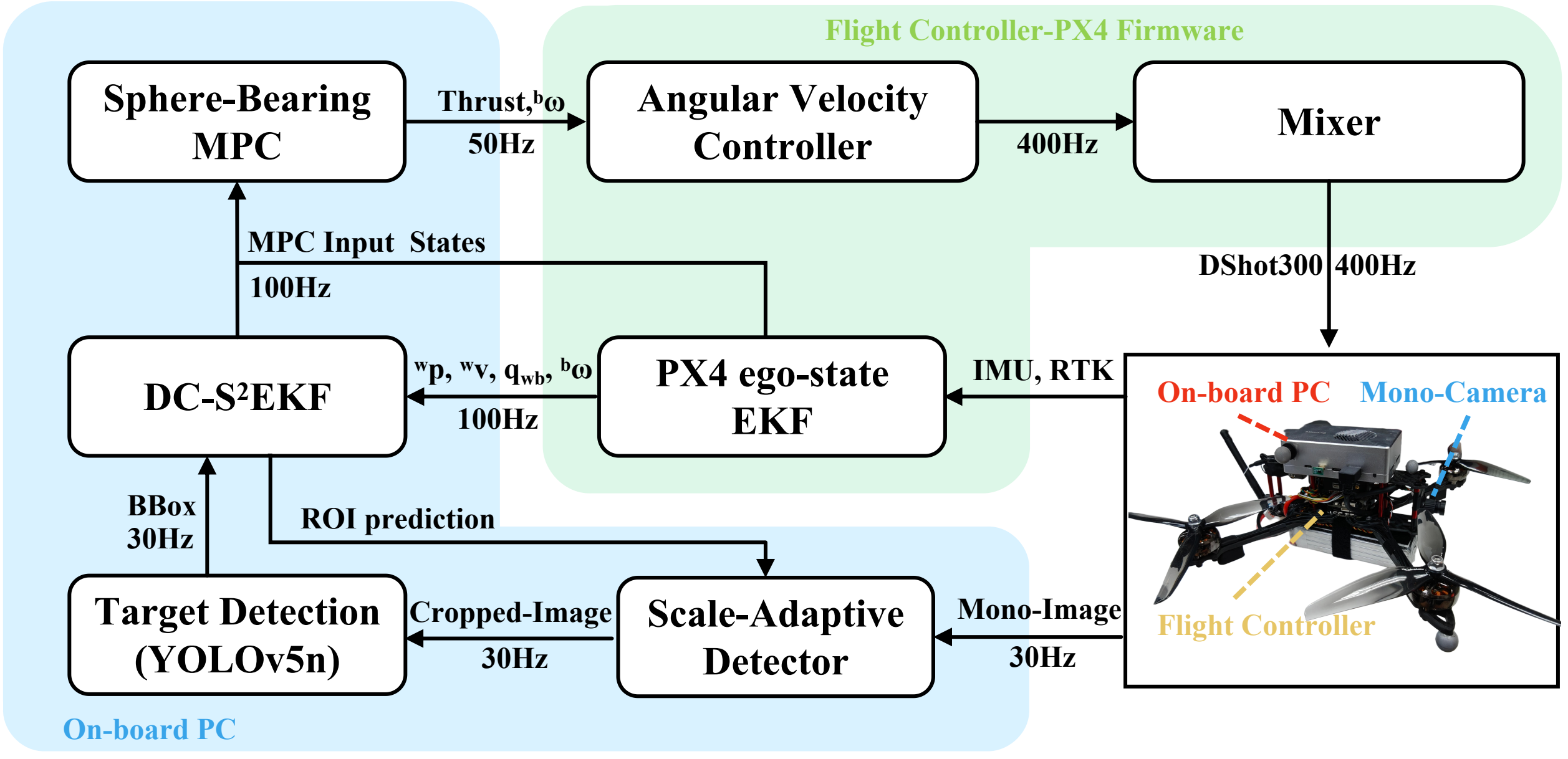

Flight Controller-PX4 Firmware
Sphere-Bearing MPC
Thrust,ᵇω
50Hz
Angular Velocity Controller
400Hz
Mixer
MPC Input States
100Hz
DShot300 400Hz
DC-S²EKF
ʷp, ʷv, q_wb, ᵇω
100Hz
PX4 ego-state EKF
IMU, RTK
On-board PC
Mono-Camera
Flight Controller
BBox
30Hz
ROI prediction
Target Detection (YOLOv5n)
Cropped-Image
30Hz
Scale-Adaptive Detector
Mono-Image
30Hz
On-board PC